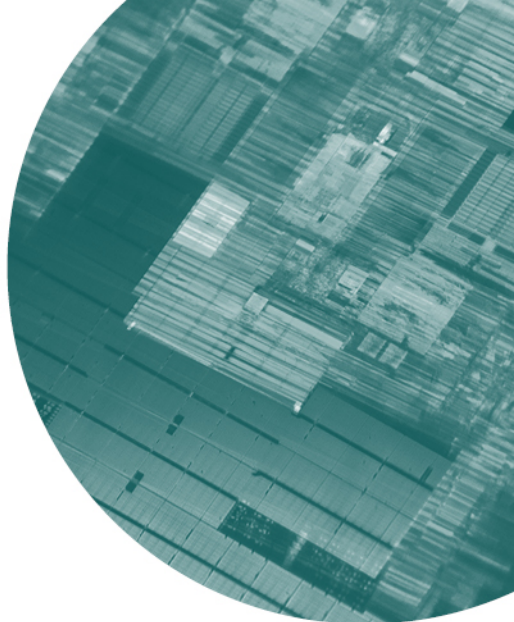




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Fabrikation von Erkenntnis

Experimente in den Digital Humanities



Experiments in the digital laboratory
What the Computational Humanities can learn about their definition
and terminology from the History of Science

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Fabrikation von Erkenntnis: Experimente in den Digital Humanities
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Melusina Press 2021

Published in **Melusina Press**, 2022

11, Porte des Sciences

L-4366 Esch-sur-Alzette

<https://www.melusinapress.lu>

Management: Niels-Oliver Walkowski, Johannes Pause

Copyediting: Carolyn Knaup, Niels-Oliver Walkowski

Cover and Layout: Valentin Henning, Erik Seitz

The digital version of this publication is freely available at

<https://www.melusinapress.lu>.

The PDF and the master copy are generated with the help of princeXML.

Print and Distribution is carried out by BoD - Books on Demand, Norderstedt

Bibliographic information of the National Library of Luxembourg: The National Library of Luxembourg lists this publication in the Luxembourg National Bibliography; detailed bibliographic data are available on the Internet at bnl.public.lu.

ISBN (Online): 978-2-919815-25-8

DOI (Publication): 10.26298/melusina.8f8w-y749-eitd

DOI (Volume): 10.26298/melusina.8f8w-y749

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Introduction

Experiment is mentioned so regularly in the Digital Humanities that one could think everything has been said about it.¹ Yet it is difficult to find serious attempts at giving a useful definition which is not too vague to be applicable.² This not only leaves the field vulnerable to criticism regarding a metaphorical appropriation of scientific terminology for potentially questionable reasons but also for technocratic speculations on how experiment could supposedly rid the Humanities of their allegedly flawed qualitative tradition by undergoing a process of making them ›more scientific‹. What we need is precise terminology and a sensible definition of where and how experimental methods can be integrated into traditional Humanities hermeneutics. This paper is an attempt at retracing the history of laboratories and experimentation as well as discussing criteria for labelling certain practices as experiments in the Computational Humanities. It makes the argument that a stricter terminological distinction between playful experimentation with the purpose of exploring data and experiment according to scientific definitions should be made.

Quantitative approaches are often discussed in conjunction with a trope of novelty and in connection to computerization. But a number of recent studies have stressed the fact that quantitative methods are neither new nor unusual in literary studies or historical research – they even predate the success story of the Digital Humanities.³ However, it seems to have been overlooked in the discussion of experimental and quantitative methods in the Digital Humanities so far that experimental methods, too, have a long-standing tradition in Humanities research in disciplines such as Experimental Archaeology or the Experimental History of Science. Discussions of experiments in the Digital Humanities rarely take into account that there are Humanities disciplines which have already integrated experimental methods into their traditional framework of hermeneutics. We can look to those disciplines for a sensible framework of experimentation in the Computational Humanities. This paper suggests that we should adopt the distinction between ›experiencing‹ and ›experimenting‹ which has become commonplace in Experimental Archaeology.⁴ It argues further that a stricter definition of experiment should be limited to a narrow subset of problems in the Computational Humanities, more specifically that which Piotrowski and Fafinski have described as Applied Computational Humanities.⁵ A rigorous definition of experiment seems impossible for the

1 Despite this being unusual in English, discipline names will be capitalized throughout this article. This facilitates differentiating them from other mentions of, for instance, the word ‘science’.

2 For a review of the rather vague attempts at defining experiment in the Digital Humanities see the contribution by Tessa Gengnagel in this volume.

3 There is a long-standing tradition of quantification in economic history cf. Piotrowski / Fafinski 2020, as well as in quantitative literary studies cf. Bernhart 2018, Buchner et al. 2020. On the fallacy of quantitative methods making Humanities more ‘scientific’ see Lauer 2020 as well as the contribution by Tessa Gengnagel in this volume.

4 To give just one example, ›experiencing‹ in Experimental Archaeology might entail reenacting a historical battle without a clear goal in mind other than, well, experiencing it. ›Experimenting‹ on the other hand could involve wanting to know at what speed a specific type of arrow pierces through armour, building an experimental setup and conducting the experiment while measuring and documenting the results.

Humanities in general which is why formulating one for the Digital Humanities as a broader umbrella field wouldn't transcend a metaphorical notion of experimentation.⁶

This paper discusses how the definitions of historical experiment and re-enactment stemming from the History of Science and more specifically, created in the research tradition of the Experimental History of Science, can be applied and related to Computational Humanities.⁷ It draws on the history of laboratory and experiment to lay out a viable approach for defining experiment in the Digital, and especially, the Computational Humanities. In order to do so, it traces back the historical origins of laboratory and experiment, drawing conclusions for the use of the terms ›laboratory‹ and ›experiment‹ with regard to datafication and quantification in the Humanities. Van Zundert asserts that the »exploration of the hermeneutic potential of computation is a challenge for the digital humanities [...]«. «⁸ Establishing the role and epistemological makeup of experiment in the Computational Humanities can contribute important insights to this endeavour.

This paper does not wish to discuss the question of how to best define the term Computational Humanities or differentiate Computational Humanities from the umbrella term Digital Humanities.⁹ Nevertheless, it makes sense to make this distinction with regard to the problem at hand. For the time being, we mean by Computational Humanities roughly those parts of Digital Humanities which are primarily concerned

5 Piotrowski / Fafinski 2020.

6 A summary of possible definitions of experiment in a more metaphorical sense of the term is given in the contribution by Tessa Gengnagel in this volume; see also Berg 2015, König 2015.

7 On the Experimental History of Science with a special regard to alchemy and chymistry: Principe 1987, 2000; Gelius 1997; Newman / Principe 2003; Reardon 2011; Fors / Principe / Sibum 2016; Neven 2016; Moureau / Thomas 2016; Hagendijk 2018; Martelli 2017; Taape / Smith / Uchacz 2020; Hendriksen 2020; Hendriksen / Verwaal 2020; Hagendijk et al. 2020; Sibum 2020; Dupré et al. 2020b, 2020a. While the false dichotomy between alchemy and chemistry has been debunked in the historiography of alchemy over twenty years ago, this artificial caesura of a Scientific or Chemical Revolution is still present in the popular imagination of alchemy, see Newman 1996; Newman / Principe 1998; Principe / Newman 2001. To rid alchemy of this connotation, the pioneers of the so-called ›New Historiography of Alchemy‹, William Newman and Lawrence Principe have suggested we should refer to early modern alchemical practice as chymistry as this was the term employed by the historical actors to refer to themselves. However, notions of alchemy and chemistry were still used interchangeably at that time. The defamation of alchemy as pre-scientific or a pseudoscience was progressively introduced towards the end of the seventeenth century by chymists themselves through a rhetoric of a supposedly new scientific openness which became prevalent and fashioned chemistry as freeing itself of the secretive language (or rather the rhetoric of secrecy) typically associated with earlier alchemical pursuits. Thinking that the scientific laboratory was born only with the rise of chemistry is thus an anachronistic assumption and would be incorrect.

8 Zundert 2016, 344.

9 On the distinction between Digital and Computational Humanities, for example: Berry 2011; Hall 2013; McGillivray / Wilson / Blanke 2018; Piotrowski 2019a, 2019b; Roth 2019; Piotrowski / Fafinski 2020; Piotrowski / Neuwirth 2020. I have pointed out potentially problematic aspects of the term ›Computational Humanities‹ in a blog post (Lang 2020). However, these aspects are not relevant to the question at hand which is why this term shall be used for lack of a better alternative to describe the subset of particularly ›algorithmic‹ research questions in the Digital Humanities. The publications cited before don't really agree with each other's definitions of Computational Humanities other than the fact that most of them recognize the mainly ›algorithmic‹ nature of the research as a common denominator.

with quantitative approaches (seen as a sub-field), distinguishing ›numerical humanities‹ which use new algorithmic methods which cannot fully be grasped using traditional Humanities hermeneutics from ›digitized humanities‹ which are concerned mainly with presentation, publication and access to data – which Michael Piotrowski has defined as ›public humanities‹.¹⁰ This article argues that speaking of experiments only really makes sense in this sub-field of the Digital Humanities which we refer to by the name of Computational Humanities. In most other cases, a rigorous definition of experiment is not possible. One of the main arguments of this paper is that we should distinguish terminologically between metaphorical experiments, such as in the case where ›experiment‹ refers to playful exploration of data, and experiments which can meet the criteria of a stricter definition of experiment which shall be given in this article. Consequently, open questions are discussed such as the hermeneutical implications of experimental methods and the implementation of practical workflows for documenting data experiments.

A history of the laboratory

The laboratory has become probably the most common metaphor for scientific as well as experimental knowledge-production.¹¹ As the birthplace of laboratories all around the world, and thus, by extension, also for the Digital Humanities Lab, the alchemical laboratory is a good starting point for the exploration of the meanings and potentials of the laboratory and experiment metaphors.¹² Hannaway states that »indications are that the laboratory was at first linked exclusively with alchemy and chemistry; only gradually, it seems, was the term extended to describe all those distinctive places where the manipulative investigation of natural phenomena was carried out.«¹³ With the emergence of Laboratory Studies in the 1980s¹⁴, the study of the laboratory became popularized but also took a decidedly sociological turn, marked by the anthropological studies in the tradition of Latour and Knorr-Cetina.¹⁵

If we are to negotiate the place and epistemological status of experimentation in the Computational Humanities, we must first reach an understanding of the terms involved, namely laboratory and experiment. While the Digital Humanities discourse is filled with the metaphors of experiments and laboratories, sound definitions about what constitutes those experimental methodologies and their place in the traditional

10 Roth 2019; Piotrowski / Fafinski 2020, 171–72.

11 On the development of the laboratory more generally, see: Ganzenmüller 1967; Hannaway 1986; Shapin 1988, 1989; James 1989; Ophir and Shapin 1991; Joly 1992; Shackelford 1993; Findlen 1994; Newman 1999; Hartung 2006; Kohler 2008; Lehmbrock 2013; Morris 2015; Black 2017.

12 A discussion on the alchemical laboratory was started by Hannaway, with a rebuttal by Shackelford following shortly after, cf. Hannaway 1986; Shackelford 1993; Newman 1999. The current state of research on alchemical laboratories is summarized in Lang 2022.

13 Hannaway 1986, 585.

14 For example: Cetina 1995; on laboratories in the alchemical context: Hannaway 1986; Crosland 2005; Martelli 2011.

15 On Laboratory Studies: Lynch 1985; Cetina 1981, 1992, 1999, 2001; Vinck 2007; Doing 2008.

hermeneutics of the Humanities are still lacking.¹⁶ This isn't all that surprising given the fact that the exact origin of the term ›laboratory‹ as we now use it remains quite obscure: it seems to appear in the modern meaning since the 16th century at the earliest.¹⁷ Furthermore, indications are that *laboratorium* »referred almost exclusively to a room or house where chemical operations such as distillation, combustion, and dissolution were performed.«¹⁸ It must be emphasized, however, that there is a significant »shared material culture of the pharmaceutical and academic-chemical laboratory«¹⁹ and some argue that we cannot speak of a laboratory in today's sense before the pre-industrial age, as »patronage of laboratories is strongly linked to industrialization« and »laboratories are very much a product of, and a symbol of, modern industrial society.«²⁰ While even the first institutionalized chymical laboratories still »resembled the workshops of apothecaries, metalworkers, and pigment makers [...] and shared many of its components with the workplaces of metal smelters, glassmakers,«²¹ the term ›laboratory‹ had acquired the meaning we now associate with it today towards the end of the seventeenth century: From this point forward, laboratories were seen as »the hallmarks of the new science – the site where theories and hypotheses were purportedly tested by experiment and from which new discoveries and useful knowledge emerged. When the word was first used, the spaces referred to were workshops rather than ›sites of science.«²²

The term *laboratorium* can already be found before the 16th century, its ending in *-orium* reminiscent of room names in monastic contexts. However, it seems that it had not yet acquired its connotation of being a place of science back then.²³ The increasing fragmentation of political power during the Renaissance period provided new opportunities which

encouraged artisans of all kinds (but especially gunners, fortification builders, architectural writers, machine builders, and alchemists) to write about their techniques, in part to advertise their know-how. [...] Craftsmen [...] wrote treatises in which they both advertised their skills and attempted to provide a theoretical elaboration of their craft experience. The publication of replicable processes and techniques would become a hallmark of laboratory activity by the mid-seventeenth century [...].²⁴

It thus seems that the turning point for the scientification of experiment came from a financial impetus: The imperative for entrepreneurial artisans to transmit their knowl-

16 On the laboratory metaphor in the DH: Siemens / Siemens 2012; Earhart 2015; Lane 2017; Pawlicka-Deger 2020; Khatib et al. 2020.

17 Especially footnote 1 in Hannaway 1986, 585.

18 Klein 2008, 769.

19 Klein 2008, 777.

20 James 1989, 1, 2.

21 Smith 2006, 292.

22 Smith 2006, 292.

23 Morris 2015, 19–20.

24 Smith 2006, 297.

edge textually and explain their theories to advertise themselves as competent practitioners. Combined with a vibrant tradition of ›practical exegesis‹ and commentary in the history of alchemy, showing off one's ability to explain the chymical theory behind one's *experimenta* became a standard requirement to gain funding and patronage.²⁵ After this shift, »the appearance of the laboratory is indicative of a new mode of scientific inquiry, one that involves the observation and manipulation of nature by means of specialized instruments, techniques, and apparatuses that require manual skills as well as conceptual knowledge for their construction and deployment.«²⁶

Fast forward to the 20th century where Science and Technology Studies (STS) began to perform ethnographical studies in the 1970s, conceptualizing the laboratory as »a gateway for understanding how scientific knowledge is produced.«²⁷ This is around the same time when first ›labs‹ for Humanities subjects started appearing during the 1980s which were »introduced as the infrastructure for facilitating and propelling the advancement of DH« as part of »the first wave of the ›infrastructure turn‹ in the humanities.«²⁸ Sociological studies emphasizing the role of the laboratory as a place of knowledge-making might be the reason why the laboratory metaphor started to become fashionable in the Humanities where Rheinberger's term of the ›experimental system‹ became a common point of reference for linking Humanities scholarship to the notion of experiment.²⁹ The metaphors of laboratory and experiment have been the subject of a plethora of turns and research paradigms in the Humanities: Be that the *cultural* and *spatial turns* related to the sociological tradition of laboratory studies³⁰, the *material turn* which has been said to have been the catalyst for the Experimental History of Science³¹, the *practical turn* equally associated with the emergence of experimental methods in historical research³² or, last but not least, in the case of the Digital Humanities, the *computational turn*.³³ However, there has been criticism regarding the use of the metaphors of laboratory and experiment in the Humanities, suggesting it to be a rhetorical device employed to profit from the prestige and status of the sciences by associating oneself with them.³⁴ This criticism is founded in the lack of a proper definition of experiment in the Humanities. While a merely metaphorical usage of the term ›ex-

25 Rampling 2020, 63–64, 97–99, 354.

26 Hannaway 1986, 585.

27 Pawlicka-Deger 2020, 2.

28 Pawlicka-Deger 2020, 6; on DH labs see also: Earhart 2015.

29 cf. Rheinberger 2001; Lehmbrock 2013, 246.

30 cf. Lehmbrock 2013, 247.

31 Some argue that it only became popular in the early 2000s due to its link to the »material turn«. See the historical overview on the *Experimental History of Science* given in: Fors / Principe / Sibum 2016, 85.

32 cf. Rheinberger 2001, 51.

33 Hall defines the computational turn as follows: »The term computational turn has been adopted to refer to the process whereby techniques and methodologies drawn from computer science and related fields—including interactive information visualization, science visualization, image processing, geospatial representation, statistical data analysis, network analysis, and the mining, aggregation, management, and manipulation of data—are used to create new ways of approaching and understanding texts in the humanities.« in: Hall 2013, 781; however, it was first brought up in Berry 2011.

34 Lehmbrock 2013, 248; Berg 2013, 142.

periment< with possible reference to Rheinberger's ›experimental system‹ certainly exists and makes sense in the Humanities, the datafication and quantification of Humanities subjects in the Digital Humanities and especially the Computational Humanities, offer the possibility of a more stringent application of the term ›experiment.<³⁵

On the notion of experiment

The notion of experiment was a central one in the making of what we now understand as Science.³⁶ Commonly, research is required to involve at least some, if not all, of the following characteristics to count as an experiment in the strict scientific sense: Relevant variables are known and the experiment has been reduced to those which are measurable and can be purposefully manipulated and isolated. The experiment is replicable and falsifiable. Before the experiment, hypotheses need to be formed so they can be investigated during the process. The setup and steps taken should be described in detail for documentation purposes. Furthermore, one might argue that the processes at the core of the investigation need to be observable. However, it should be noted that there is no uniform definition of experiments and definitions vary from field to field.³⁷

In the digital context, we need to add digital documentation as well as long-term accessibility of both the algorithms and data used. And obviously, the phenomenon which is meant to be experimented on needs to be digitally formalizable. An important advantage of the controlled environment of an experimental setup is that variables of the phenomenon to be investigated can be purposefully manipulated which makes it easier to determine cause and effect using experiment than in the case of mere observation. Experiments also came to be a central aspect in demonstration and teaching. This specific case of experimenting would usually be referred to as ›replication‹ which

generally means the repetition of an experiment in order to check or confirm prior results. Additionally, in educational contexts students perform canonical experiments in exercises that are often considered to be replications. In that case, successful repetition of canonical experiments is a means of calibrating the mind of the student, generally with the aim of proving particular scientific claims as stated in textbooks. When historians rework or reproduce a process or an experiment as a historiographical tool they are not replicating in these scientific or pedagogical senses, but are instead seeking fresh historical information.³⁸

The issue of replication and replicability is one which has come under scrutiny as of late in connection to a methodological ›replication crisis‹ and its relevance for the Computational Humanities will be discussed in more detail later in the article.³⁹

35 On Rheinberger's ›experimental systems‹ see for example: Rheinberger 2001.

36 On experimentation, see: Klein 2008; Breidbach et al. 2010; Berg 2013. On the role of experiment in empiricism see Berg 2015.

37 Berg 2013, 141.

38 Fors / Principe / Sibum 2016, 93.

In the early modern period, *experimenta* became vehicles for the deliberate production of *experientia*, i. e. experience of natural processes the observer was partaking in.⁴⁰ In a dictionary of alchemical specialist vocabulary, experiment is described as both foundation and touchstone for hypotheses.⁴¹ This warrants caution for it mixes up two distinct parts of the research process, that is the formation of hypotheses and the rigorous testing thereof by means of experiment. It shall be argued here that it is crucial to carefully distinguish these different steps involved in scientific knowledge-making practices surrounding experimentation. When it comes to terminology, the Digital and Computational Humanities can draw on fields which have already developed it, such as the Experimental History of Science or Experimental Archaeology. Due to it being based mainly on material objects as evidence, Experimental Archaeology is a less useful comparison here than the Experimental History of Science which has material aspects but is still mostly a field based on textual sources.

The Alchemy of Experience

Despite some earlier pioneering works, the Experimental History of Science is a relatively young subfield which has only gained traction since the ›experimental turn‹ in the History of Science, starting in the early 2000s.⁴² In the meantime, the use of experimental methods in the historiography of alchemy, pioneered especially by Lawrence Principe, has resulted in tremendous revisions of the current state of the art in the field, so much so that the journal *Science* spoke of an ›Alchemical Revolution.‹⁴³ A whole range of experimental methods has become established in the Experimental History of Science under the label of ›RRR methods‹ (reconstruction, replication, re-enactment).⁴⁴ In the context of the ›RRR methods‹, the distinction between ›experiencing‹ and ›experimenting‹ is emphasized, with ›experimenting‹ implying a stronger commitment to the controlled and deliberate testing of hypotheses through experiments.⁴⁵ Like mentioned above, early modern laboratories as the sites of experimentation are characterized as spaces used for the controlled manipulation of one's material of study with the aim of generating questions, forming hypotheses and linking the observed to theoretical ideas. ›Experiencing data‹, so to speak, for the purpose of testing hypotheses as well as the isolation of specific features so that cause and effect can be determined.

How can Computational Humanities research be framed as experiment? It seems that it is on the basis of quantitative evidence and manipulation thereof that the notion of experiment hinges in the context of the Computational Humanities.⁴⁶ However, the

39 Replication in the context of the Humanities has been discussed in Peels / Bouter 2018 and Peels 2019.

40 Berg 2013, 141.

41 Barke 1991, 232–33.

42 Fors / Principe / Sibum 2016, 85.

43 cf. Reardon 2011.

44 Hendriksen provides a definition: »Performative methods include, but are not limited to, reconstruction, replication, and re-enactment (RRR) of historical experiments, apparatus, processes, and techniques« (Hendriksen 2020, 314).

45 cf. Fors / Principe / Sibum 2016, 93; Hendriksen 2020, 317.

seminal 1966 publication on *Quantification in History* is quite reserved with regard to the scope of applicability of quantitative methods in the Humanities, stating »that counting, when circumstances permit it, may assist in the explanation of a limited class of historical problems.«⁴⁷ Some like to frame experimentation as infallible and utterly neutral if done right, sometimes transferring this supposed epistemological prowess to science – the field that experiment is symbolic of – as well as to quantification which seems fundamentally related. Yet there is no way of getting around the qualitative aspect inherent in and pivotal to experimentation when interpreting results of experiments to draw conclusions from them: »the hypotheses used to connect observation and theory are, no matter how plausible they at first appear, always open to challenge.«⁴⁸ The idea of introducing qualitative elements or intuition into experiment seems to be viewed with a certain disdain when, for example, McGillivray et al. demand that we should »exclude intuition as inadmissible as evidence«⁴⁹ but acknowledge that it *can* be part of experimental research in the process of forming hypotheses: »A *hypothesis* originates from previous research, intuition, or logical arguments, and is »a claim that can be tested empirically, through statistical hypothesis testing on corpus data« (Jenset and McGillivray, 2017, 42).«⁵⁰ Sinclair and Rockwell bring up the notion of experimentation in the context of quantitative text analysis when they state that »the idea that text analysis and visualization are interpretative practices may seem paradoxical at first glance, since the digital is founded on matching and counting, but no amount of counting can produce meaning. On the other hand, digital tools do facilitate experimentation with the representation of digital texts [...].«⁵¹

But what exactly do they mean by experimentation in this context? Surely it is not the rigorous testing of hypotheses a strict definition of experiment would imply. Jockers, on the other hand, argues that »[c]omputational analysis may be seen as an alternative methodology for the discovery and the gathering of facts. [...] The computer is a tool that assists in the identification and compilation of evidence. We must, in turn, interpret and explain that derivative data.«⁵² This is the pre-experimental phase McGillivray et al. concede to intuition. There can hardly be experimentation without exploration. In our previously introduced framework of experience and experiment, a hypothesis defined as above is a possible result of experience and can later be used in experimentation for verification or falsification purposes. In a Computer Linguistics publication, McGillivray et al.

46 Regarding quantification in (digital) history, see for example: Aydelotte 1966; Brendecke 2015; Vogeler 2015; Piotrowski 2019b.

47 Aydelotte 1966, 803.

48 Aydelotte 1966, 818.

49 McGillivray / Wilson / Blanke 2018, 54.

50 McGillivray / Wilson / Blanke 2018, 54.

51 Sinclair / Rockwell 2016a, 288.

52 Jockers 2013, 30.

define *evidence* in quantitative historical linguistics as the set of »facts or properties that can be observed, independently accessed, or verified by other researchers« (Jenset and McGillivray, 2017, 39).

[...] *Quantitative evidence* is »based on numerical or probabilistic observation or inference« (Jenset and McGillivray, 2017, 39), and the quantification should be independently verifiable.⁵³

Earlier, the notion of metaphorical laboratories was brought up with regard to Digital Humanities Labs. While it is quite clear that the term ›laboratory‹ as used in Digital Humanities context is often a metaphorical usage, following the sociological and anthropological tradition of Laboratory Studies, not all experiments in the Digital Humanities are purely metaphorical ones. In fact, it shall be argued here that it is crucial for the Computational Humanities to make a sharp distinction between different terms referring to experimental methods, as both Experimental Archaeology and the Experimental History of Science are adamant about for good reason. I argue that in order to fully grasp the meaning of experiment in the Computational Humanities, we should include the early modern concept of *experientia*, i. e. experience, back into our terminology. Not all experiments in the Digital and Computational Humanities can satisfy the criteria for experimentation in the scientific sense. They, however, also aren't experiments in the metaphorical sense only. Making this distinction is more than merely a matter of being politically correct. Due to the rigorous standards it implies, experiment holds a different epistemological status than mere experimentation in the sense of *experientia*. Oftentimes, we merely try to ›experience‹ in a playful way for purposes of data exploration rather than engaging in rigorous testing. This is valid, yet should not be confused terminologically with experiment in the strict scientific sense. Oftentimes, we experiment »not to get results but to generate questions [...]». Ramsay (2014) calls this the screwmenneutical imperative.«⁵⁴ Confirmative analysis (*experiment*) serves to test hypotheses or to falsify them, whereas explorative analysis (*experientia*) tries to generate hitherto unseen ›views‹ on data, on both the micro- and macroscopic scale, pointing to potentially relevant features or patterns.⁵⁵

Earlier it has been argued that the definition of a strict notion of experiment only makes sense in the subfield of questions which scholars have labelled as the Computational Humanities. However, when it comes to a rigorous definition of experimentation, we need to go even further. Not unlike in chemistry, we can only label something as an experiment in the strict sense when it involves the manipulation and observation of the subject matter. This is only the case in the subfield which Piotrowski and Fafinski have summarized under the title of Applied Computational Humanities, i.e. those cases in which computational methods are used to answer Humanities research questions.⁵⁶ Epistemically, these are in stark contrast to research pertaining to Theoretical Compu-

53 McGillivray / Wilson / Blanke 2018, 54.

54 Sinclair / Rockwell 2016a, 288.

55 cf. Rehbein 2017, 332.

56 Piotrowski / Fafinski 2020, 177.

tational Humanities which focuses on improving algorithms or benchmarking. Such work cannot be labelled experimental in the strict sense because it does not have a hypothesis which is being investigated systematically. Improving algorithms primarily concerns the experimental setup, not a subject matter.

However, some questions remain. For instance, can we speak of experiments in the case of algorithms which function as black boxes? The criterion of manipulating a subject matter is fulfilled because the parameters can be tweaked. Still, the process in itself is not observable. An additional issue with many algorithms is the fact that they include elements of randomization which might lessen the degree of replicability depending on what the goal of the experiment is. Here, procedures should be developed for making standardized judgements with regard to the experimental setup's ability to reproduce similar effects even if the exact experiment cannot be replicated.

Experimental methods, hermeneutics and the need for replicability

The place of experiment in Humanities hermeneutics

Now that we have discussed what a definition of experiment in the Computational Humanities could look like, we need to decide on the place of experimental methods in the hermeneutic framework of the Humanities. The Experimental History of Science has been using experimental methods for quite some time and thus, has had time to create definitions and terminologies.⁵⁷ Despite the fact that actual scientific experiments on physical matter have different implications than data experiments in the Digital Humanities, there is a surprising overlap in the potential hermeneutic problems encountered. The discussion on the hermeneutics of those experimental methods in the Experimental History of Science has characterized them as providers of (sensual) data which can be used to fill in the <documentary gaps> in historical sources.⁵⁸ Both the methods of the Experimental History of Science as well as the Computational Humanities are to be seen as complementary to traditional scholarship. Much like methods for quantitative text analysis or *Distant Reading*, such experimental methods are understood as supplementary sources, providing new data for historical contextualization, in cases where *close reading* alone does not lead to usable interpretations, mostly due to a high degree of uncertainty and vagueness in the historical sources.⁵⁹ They both do not per-

57 On the laboratory and experiment in the History of Science, especially the Experimental History of Science: Hannaway 1986; Cetina 1995; Dupré et al. 2020b, 2020a; Hagendijk et al. 2020; Sibum 2020.

58 Fors / Principe / Sibum 2016, 91.

59 The role of *distant reading* is not to replace *close reading*, as Moretti had initially promoted, but to inform a potential reader of pieces of text worth studying in detail, of zooming into the smallest details, but also keeping enough overview over the corpus as a whole to contextualize them. It is a hermeneutical tool for the study of texts to enrich traditional approaches to textual analysis rather than a method to replace them; cf. Moretti 2013; Jockers 2013; Sinclair / Rockwell 2016b.

tain to the interpretation of historical sources but much rather make up part of the ›evidence gathering‹ which precedes interpretation.⁶⁰ They serve to quantify historians intuitions and hypotheses;⁶¹ but in the end, an experimental approach »provides *data*, not interpretation.«⁶² Jennifer Rampling has referred to this use of experiment for text interpretation as ›practical exegesis.‹⁶³ When experimental methods are being used to investigate texts, we might consider this »a new kind of philological tool, in the form of [...] experimental practice.«⁶⁴

By integrating experimental methods into the arsenal of traditional hermeneutics, we encounter the problem that the types of knowledge involved in practical processes such as *tacit* or *gestural knowledge* are particularly difficult to encode textually. In the History of Science, a whole sub-field has emerged investigating artisanal knowledge-making.⁶⁵ Inexplicit forms of knowledge are challenging as a type of evidence to be used by computers which are excellent and easy-to-use devices only when it comes to *explicit* knowledge.⁶⁶

Documenting experiments in a replicable fashion

However fruitful experimental analyses can be, the problem of their adequate documentation needs to be addressed. How can all the tacit and gestural knowledge involved in them be encoded, if it can be made explicit at all? In the very least, for experiments to be replicable, all data used, the algorithms and / or software in the current version need to be made accessible. As it has been lamented regarding the documentation of historical experiments, »we need to rethink our models for publishing and crediting, as well as the ways in which we store and disseminate data« stemming from digital experiments in a sustainable manner.⁶⁷ Content models for such research data must be created and incorporated into the ›tools of the trade‹ for the Digital Humanities.

While this paper has proposed terminology for talking about experiments in the Computational Humanities and suggested plausible definitions, there remains a number of open questions regarding best practices for publishing the results of research containing experimental methods. Mareike König has brought up the question of

60 Jockers 2013, 10. The possibility (or usefulness) of generalizations in the Humanities which might be made using computational methods is discussed in Piper 2020 but this is not in the scope of the subject of the present article.

61 cf. Sinclair / Rockwell 2016a, 282.

62 Moretti 2005, 9.

63 Rampling 2020, 63–64, 97–99, 354.

64 Rampling 2020, 98.

65 Prominently, the public humanities project ›Making and Knowing‹ among many others: Smith 2006, 2017, 2020; Nummedal 2011; Neven 2014, 2016a, 2016b; Principe 2014; Dupré 2017; Valleriani 2017; Hagendijk 2018; Hendriksen / Verwaal 2020.

66 Polanyi 2009; Collins 2010; on gestural knowledge: »[...] practical engagement also builds up a working knowledge in the practitioner that Sibum has called ›gestural knowledge‹. [...] The nature of this gestural knowledge is well-described in his difficult-to-translate German phrase ›das in seinem Handlungsvollzug gebundene Wissen‹« (Fors / Principe / Sibum 2016, 92; Sibum 2020).

67 Hendriksen 2020, 317.

whether the Humanities too need to have a discourse surrounding the ethics of experiment.⁶⁸ In the case of experiments in the realm of Applied Computational Humanities as described in this article, this would seem to pertain mostly to ensuring good scientific practice and, in the case of recent data, potential privacy issues. In the future, replication – or at least testing if experiments are documented in a way which would theoretically allow replication, i.e. ensuring replicability of Computational Humanities research – should be included into the peer review process.⁶⁹ Best practices should be formulated and content models for the publication of replicable experiments should be developed. Meta studies should be done to ascertain to which degree existing Computational Humanities research is already replicable: Do publications include sufficient information that the experiment could be repeated or at least the effect observed could be reproduced using a similar data set? Making digital experiments replicable includes archiving or versioning the state of any software or datasets used in the process as well as detailed documentation and metadata on how the experiment was conducted or how the setup was devised.

Another issue to take into account in the realm of machine learning especially is the presence of randomization in algorithms. Here the results might never be exactly the same even if an exact replication is performed. How can we establish a threshold for plausibility checking results even if the exact results cannot be replicated? At which point do we accept that an observed effect has been reproduced sufficiently for the experiment's results to be considered valid? Another area of discussion should be that of statistical interpretation: Do we agree with conclusions drawn from the data which has resulted from experiment? Who vouches for the integrity of the results of statistics-heavy research and conclusions drawn from it? These skills don't usually make up the core competences of scholars trained in the Humanities. Computational Humanities publication venues might want to include statistics experts into peer review processes to ensure that statistical misinterpretations or misrepresentations of results derived from analysing tabular or statistical data by means of text or data science are caught early on in the process.

Providing templates for the documentation and discussion of experimental results in the Computational Humanities might be a first step towards ensuring that all relevant information is present. However, journals might have to publish or archive the data analysed using experimental methods to ensure publicly available data doesn't get lost or changed over time. Not all digital data providers use versioning systems for their data which might very well be subject to being changed over time. All in all, the mode of publication of experimental research should make replication as easy as possible to encourage good research practices and discourage tampering with results.⁷⁰

68 König 2015.

69 Guidelines for implementing replication in the Humanities were given in Peels / Bouter 2018 and Peels 2019.

70 Data and corpus criticism is another realm relevant to this enterprise. However, it would have been too vast to be dealt with in the scope of the present article.

Determining the epistemic significance of experimental data

Something more should be said about the place we attribute to experiment in the Humanities: Given the very narrow range of useful applications of experiments in Computational Humanities it would be wrong to overstate their potential for the Digital Humanities as a whole. All Digital Humanities research does not need to become experimental. Overstating the value of experiment as a novel addition to Humanities methodologies can lead down an epistemic slippery slope which pits supposedly more objective scientific methods against the traditionally more qualitative methodology of the Humanities.⁷¹ But notwithstanding the pitfalls (and epistemic naïveté) of a rhetoric of scientification in the Humanities, in some cases experimental methods do serve to fill gaps in knowledge which could not be filled using other methods, enabling us to make better-informed interpretations. Yet in order to make use of this potential we need to be able to identify sensible use cases for experiments in Digital Humanities research. It is for this reason that this paper has stressed the need for us to come up with a clearer terminology surrounding experimentation in Digital Humanities. Scholarly inquiry, be that in the Sciences or the Humanities, ultimately always comes down to knowledge-making using modelling.⁷² Both data-driven or supposedly traditional qualitative approaches are evidence- and model-based research. Apart from how results are created, both approaches rely on data. Data is never entirely neutral. Datasets must be created and this process has its flaws. Be that because they are created as models by phenomenotechnical devices⁷³ or because corpora might not be balanced or representative. All this data shows nothing more than (numerical) patterns. In order for those patterns to be made useful, narratives need to be created based on them in the process of interpreting it.⁷⁴ It would thus be wrong to assume that quantitative research is more objective and doesn't involve qualitative aspects such as interpretation and meaning-making from numbers.⁷⁵

We have to avoid blindly falling for the tropes of novelty, exactitude and or making the Humanities research >more scientific< by means of experimental methods, especially given the fact that neither quantitative nor experimental methods are in any way new or foreign to what constitutes the Humanities. However, I have argued that an important reason why such rhetoric still flourishes to this day is the fact that we lack a comprehensive definition of what experimentation can mean in the Humanities. It was the purpose of this paper to suggest one possible definition.

71 On rhetoric surrounding experiments and laboratories in the Humanities see the contribution by Tessa Gengnagel in this volume as well as Berg 2013.

72 cf. Stachowiak 1973.

73 See forthcoming work by Michael Piotrowski.

74 Berry 2011, 14.

75 This has been discussed at length in a number of publications such as Moretti 2005, Berry 2011, Jockers 2013 or Sinclair / Rockwell 2016a.

Conclusion

In this paper, the historical origins of the terms laboratory and experiment as well as their metaphorical and non-metaphorical uses in the Humanities and Sciences were discussed and related to the usage of the term ›experiment‹ in the Computational Humanities. It has defined the role of experimental methods in Humanities hermeneutics as standing in between historical sources and their interpretation, in the phase of ›evidence gathering‹ which informs subsequent interpretations. The view sometimes expressed in the context of science that data could somehow render research more quantitative, replacing elements of qualitative interpretation and thus make research more objective is epistemically naive and a fallacy. Experiments merely serve to reduce information entropy by producing more metadata which can be used to fill in documentary gaps in cases where the historical record doesn't furnish enough information to answer a given research question.

It is safe to say that most Digital Humanities scholars disapprove of the metaphorical use of technical terms they consider as pertinent to their field, such as the term ›database‹. Even though we might not admit it openly, it is considered somewhat disrespectful. The metaphorical usage of the term ›experiment‹ in Humanities contexts has already come under scrutiny. It has been argued in this paper that the Computational Humanities should distinguish carefully between experimentation in the sense of experience (*experientia*) and experiment in the traditional scientific sense. As terminology to address digital research practices which we consider experimental, yet aren't rigorous enough to count as actual experiments, this paper has suggested we should use the notion of experience and distinguish very carefully between experience and experiment as it is being done both in the Experimental History of Science as well as in Experimental Archaeology.

We can experience data in an intuitive way as a valid part of the pre-experimental phase of the research process by manipulating it just like early modern scientists set out to study nature by experiencing and purposefully manipulating its processes. Be that as it may, workflows for the documentation of and best practices for both those types of experimental practices are still lacking and need to be established. According to the strict definition of experiment given in this paper, not a lot of research qualifies as experimental thus far. This is mostly due to the lack of institutionalized or standardized approaches for ensuring the replicability of results. Accordingly, this definition is not meant as a description of past research but rather an idealized notion of experiment inviting Computational Humanists to optimize their workflows and publication strategies. As long as reproducibility is not a given, experimental methods will not be able to deliver more the objective results they are often promised to produce. If results cannot be independently verified, bad research practice could become rampant such as the cherry-picking of evidence or even tampering with results.

While the Digital Humanities can't agree on a more or less uniform usage of the term 'experiment', it will continue to be criticized as merely performative language with the goal of earning prestige associated with the Sciences. A well thought out notion of how experimental methods figure into traditional Humanities hermeneutics might also

be able to reduce the artificial divide between supposedly qualitative, intuition-based Humanities and allegedly objective Sciences. Further theoretical reflection of experiment's epistemic role in research is much needed. This article has proposed a conceptual framework of experiments as part of the hermeneutic process of the narrow field of Applied Computational Humanities. It has been argued that having a strict definition of experiment is important to defend the field against the rhetoric of supposedly improving the Humanities by means of making it ›more scientific‹.

Despite quantitative methods currently being a more dominant research paradigm in the Humanities than Aydelotte (1966) might have predicted, they remain an epistemically narrow sub-field of Humanities research questions. Quantitative methods alone will never be able to answer the predominantly qualitative Humanities research questions. Not all those can be rendered measurable, quantifiable or computational. Thus most of them don't lend themselves to experiment in the literal sense of the word. Experimental methods are merely an addition to the methodological arsenal of Humanities hermeneutics which can be used fruitfully for the purposes of exploratory data analysis and evidence gathering to back up interpretations. It is important, however, that we realize there is room for experiments outside of the vague and metaphorical usage of the term in Digital Humanities. Yet those more rigorous definitions of experiment only apply to a fairly narrow subfield of research questions, mainly in the realm of Applied Computational Humanities. Just like we shouldn't fall for the fallacious belief that a stronger focus on experiment could somehow ›improve‹ Humanities research at its core by making it more data-driven or quantitative, we also should avoid an outright dismissal of such methods as they can be important tools for filling in epistemic gaps in the evidence gathering phase before the interpretation of historical sources.

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